

Executive Summary

A focused reviewer covering cell structure, bacterial classification, and metabolism for the upcoming examination.

KEY TAKEAWAYS

- Gram staining separates bacteria by cell wall
- Aerobic respiration yields more ATP than fermentation
- Organelles compartmentalize eukaryotic metabolism

Topics

TOPIC 1

Cell Wall & Membrane

Structural layers that define prokaryotic and eukaryotic boundaries.

Peptidoglycan Layer

- Thick in Gram-positive, thin in Gram-negative
- Cross-linked by peptide bridges

TOPIC 2

Metabolic Pathways

Energy-generating routes central to microbial growth.

Glycolysis

- Converts glucose to pyruvate
- Yields 2 ATP per molecule

Terms & Definitions

Term	Definition
Aerobe	An organism that requires oxygen for growth.
Fermentation	Anaerobic ATP generation without an electron transport chain.
Plasmid	Small circular extrachromosomal DNA.

Key Facts & Formulas

Formula / Fact	Context
ATP = C₁₀H₁₆N₅O₁₃P₃	Adenosine triphosphate
$\Delta G^\circ = -RT \ln K_{eq}$	Standard free-energy change